

THE ENDOCRINE & METABOLIC SYSTEM

The Body's Chemical Messengers

Your endocrine system is a network of glands that produces hormones—the chemical messengers that regulate nearly every function in your body. From growth and metabolism to reproduction, sleep, mood, and energy, hormones help keep your body in balance.

Because hormones operate in incredibly small amounts, even subtle disruptions have the potential to influence normal biological function. This is why scientists have become increasingly interested in studying how certain environmental chemicals, including some associated with plastics, may interact with the endocrine system.

Why the Endocrine System Matters

Unlike the heart or lungs, the endocrine system doesn't consist of a single organ. It includes the thyroid, pancreas, adrenal glands, ovaries, testes, and other hormone-producing tissues that work together to coordinate the body's most important functions.

Many plastics contain or are manufactured with chemicals such as **bisphenols (including BPA)** and **phthalates**, which are known as **endocrine-disrupting chemicals** because they can interfere with normal hormone signaling.

It is important to distinguish these chemicals from microplastics themselves. While researchers are studying the effects of microscopic plastic particles, endocrine-disrupting chemicals have been investigated for decades and have a much larger body of scientific evidence.

What the Research Has Found

Large human studies have linked exposure to certain endocrine-disrupting chemicals with changes in hormone levels, reduced fertility, earlier puberty, metabolic disorders, obesity, and type 2 diabetes. While these studies do not always prove cause and effect, the consistency of the findings has raised significant public health concerns.

Researchers are now investigating whether microplastics may also contribute to these effects by carrying endocrine-disrupting chemicals into the body or by triggering inflammation and

oxidative stress. This area of research is still developing, and many questions remain unanswered.

Scientists are also studying whether long-term exposure to these chemicals may influence thyroid function, insulin regulation, testosterone, estrogen, and other hormones that play essential roles throughout life.

What This Means

Hormones work like an orchestra. Each one must be released at the right time, in the right amount, to keep the body functioning properly.

Scientists have shown that certain chemicals associated with plastics can interfere with this delicate communication system. Exactly how much everyday exposure contributes to long-term health risks is still being investigated, but reducing unnecessary exposure has become an important focus of environmental health research.

Understanding the endocrine system reminds us that our bodies are influenced not only by the food we eat and the exercise we perform, but also by the environment we live in every day.

Key Takeaways

- The endocrine system regulates growth, metabolism, reproduction, sleep, mood, and many other essential body functions.
- Certain chemicals associated with plastics, including **BPA** and **phthalates**, are recognized endocrine disruptors and have been extensively studied.
- Human studies have linked higher exposure to some endocrine-disrupting chemicals with changes in hormone function, fertility, metabolism, and other health outcomes.
- Researchers are now investigating whether microplastics themselves may also influence the endocrine system, but this research is still evolving.
- More studies are needed to fully understand how long-term exposure to both plastic particles and endocrine-disrupting chemicals may affect human health over a lifetime.